

Decomposition of Graphs: Previsit and Postvisit Orders

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Graph Algorithms
Data Structures and Algorithms

Learning Objectives

- Compute the preorder and postorder numbers for a DFS.
- Understand why these numbers might be important.

Outline

1 Definition

2 Properties

Need to Record Data

- Plain DFS just marks all vertices as visited.
- Need to keep track of other data to be useful.
- Augment functions to store additional information.

Previsit and Postvisit Functions

Explore(v)

visited(v) $\leftarrow$ true

previsit(v)

for (v, w) $\in E$:

 if not visited(w):

 explore(w)

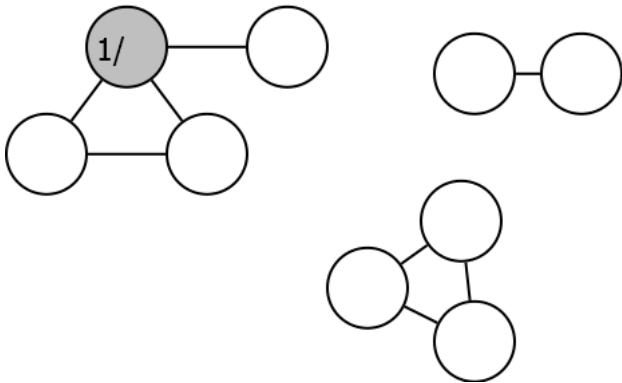
postvisit(v)

Clock

- Keep track of order of visits.
- Clock ticks at each pre-/post- visit.
- Records previsit and postvisit times for each v .

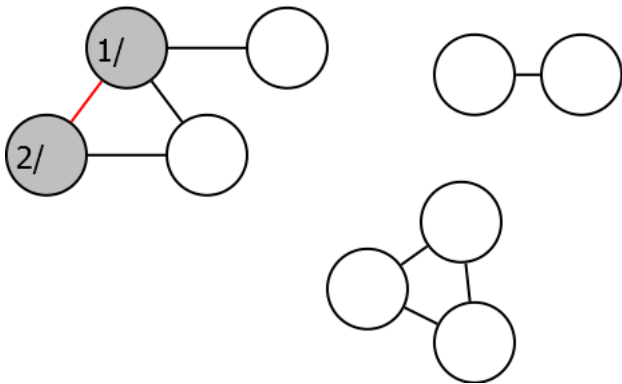
Example

Clock: 1



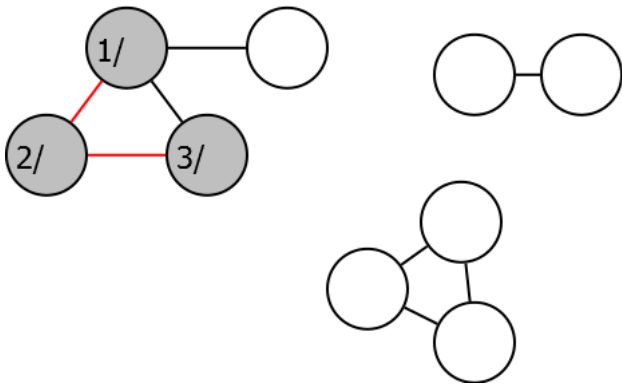
Example

Clock: 2



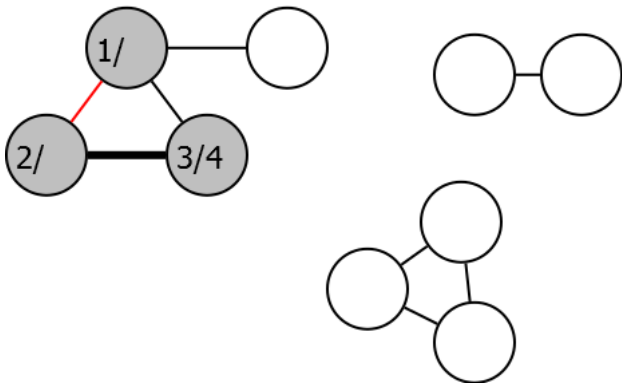
Example

Clock: 3



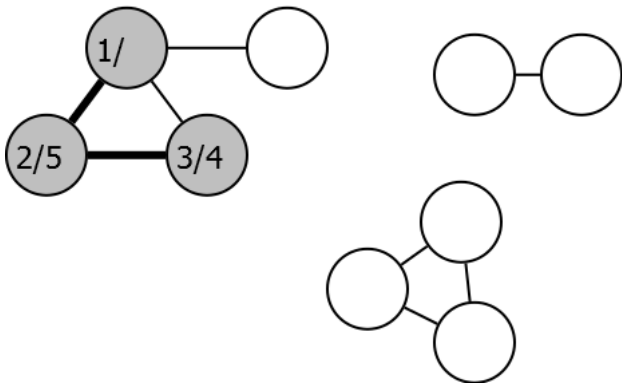
Example

Clock: 4



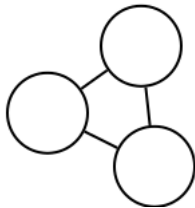
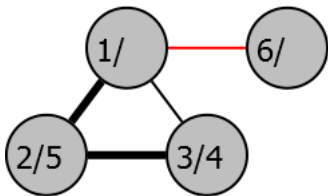
Example

Clock: 5



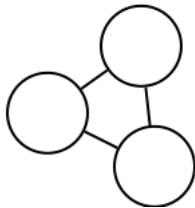
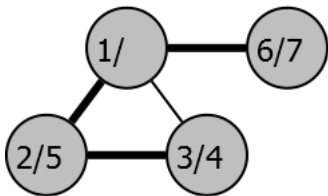
Example

Clock: 6



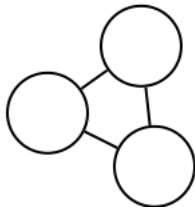
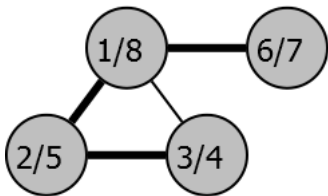
Example

Clock: 7



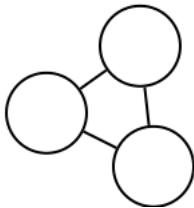
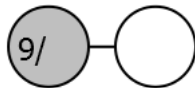
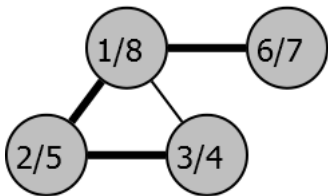
Example

Clock: 8



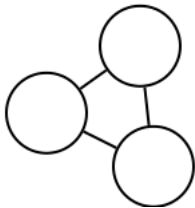
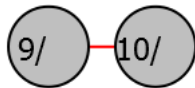
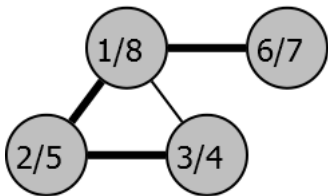
Example

Clock: 9



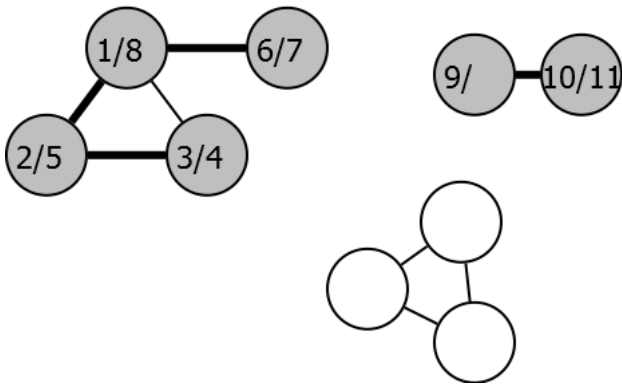
Example

Clock:10



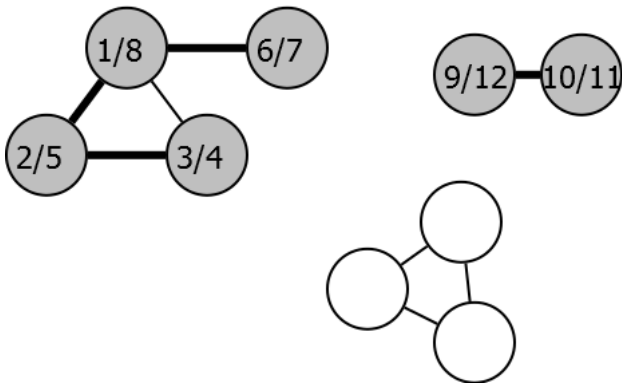
Example

Clock:11



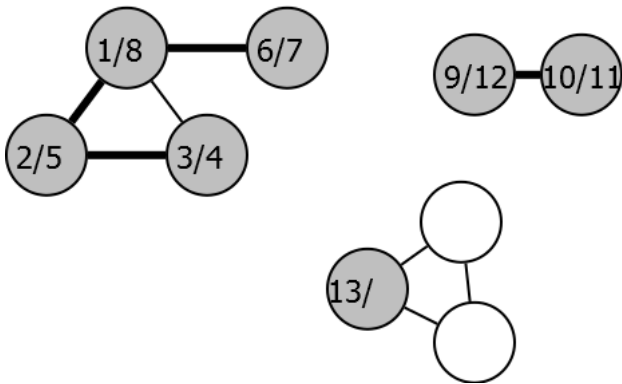
Example

Clock:12



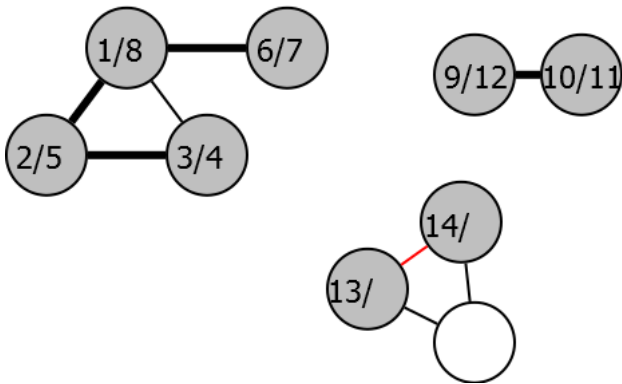
Example

Clock:13



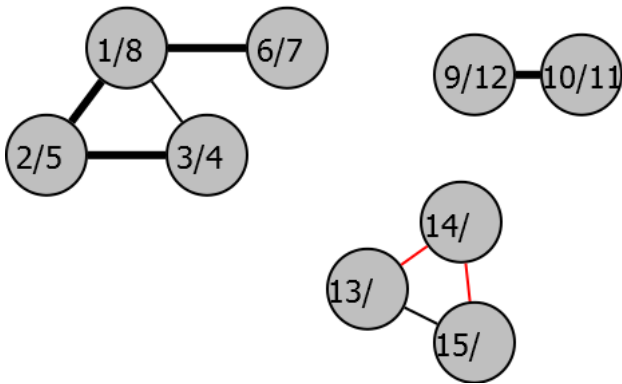
Example

Clock:14



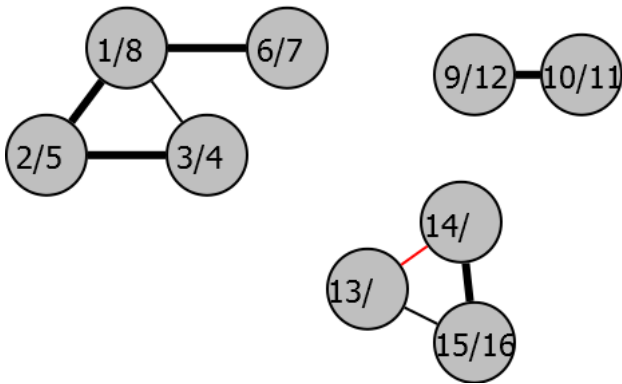
Example

Clock:15



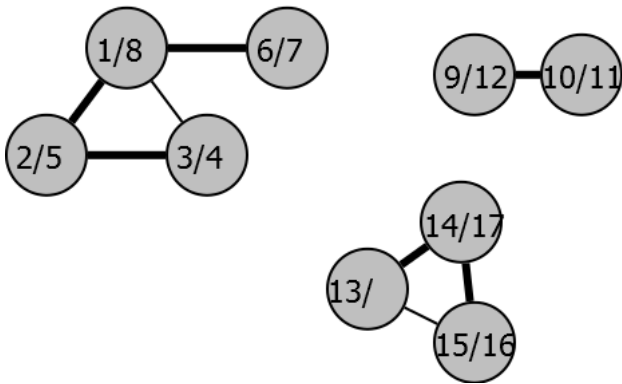
Example

Clock:16



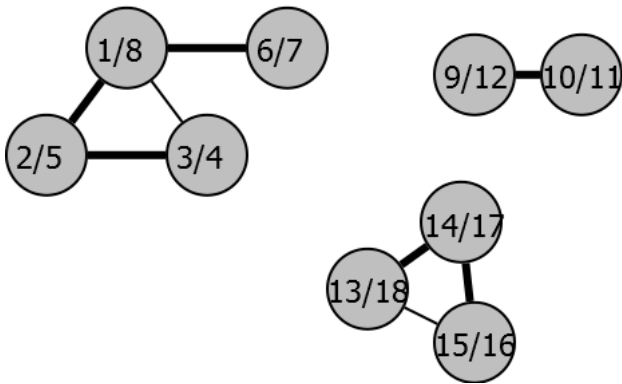
Example

Clock:17



Example

Clock:18



Computing Pre- and Post- Numbers

Initialize clock to 1.

previsit(v)

$\text{pre}(v) \leftarrow \text{clock}$

$\text{clock} \leftarrow \text{clock} + 1$

postvisit(v)

$\text{post}(v) \leftarrow \text{clock}$

$\text{clock} \leftarrow \text{clock} + 1$

Outline

1 Definition

2 Properties

Result

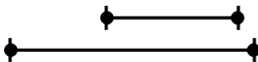
Previsit and Postvisit numbers tell us about the execution of DFS.

Lemma

For any vertices u, v the intervals $[\text{pre}(u), \text{post}(u)]$ and $[\text{pre}(v), \text{post}(v)]$ are either nested or disjoint.

Explanation

Nested

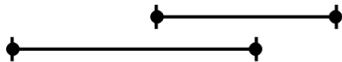


Disjoint



Explanation

Interleaved (not possible)



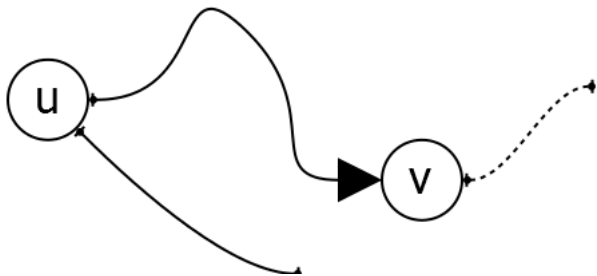
Proof

Assume that u visited before v . Two cases

- Find v while exploring u (u an ancestor of v)
- Find v after exploring u (u a cousin of v)

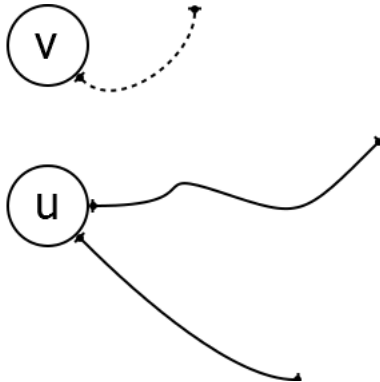
Case I

If you explore v while exploring u , cannot finish exploring u until done exploring v .
Therefore nested.



Case II

If explore v after finish exploring u ,
 $\text{post}(u) < \text{pre}(v)$, therefore disjoint.



Problem

Which of the following tables is not a valid set of pre- and post- orders?

Vert.	Pre	Post
A	1	8
B	9	10
C	3	4
D	2	7
E	5	6

Vert.	Pre	Post
A	1	9
B	8	10
C	2	7
D	3	6
E	4	5

Solution

Which of the following tables is not a valid set of pre- and post- orders?

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Vert.	Pre	Post
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E	4	5

Next Time

Directed graphs.